Adaptive Wavelet-Based Image Compression Using Bayesian Thresholding and Run-Length Encoding

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***Abstract*—This paper presents an efficient image compression framework combining wavelet transform, Bayesian thresholding, adaptive quantization, and run-length encoding. The method reduces redundancy while preserving visual quality and achieves a balance between compression ratio and reconstruction accuracy. A comparison with JPEG compression demonstrates improved performance in terms of PSNR and visual quality.**

***Index Terms*—Image compression, wavelet transform, Bayesian thresholding, quantization, run-length encoding, JPEG comparison**

I. Introduction

The rapid growth of digital images demands efficient compression techniques. Image compression reduces storage requirements while maintaining visual quality.

Earlier work such as [1] explored the use of self-organizing feature maps with wavelet transforms. While effective, such approaches involve higher computational complexity.

This work proposes a simpler hybrid model combining wavelet transform with statistical thresholding and encoding, and evaluates its performance against JPEG compression.

II. Literature Review

Wavelet-based methods enable multi-resolution representation of images. Thresholding improves sparsity, while encoding reduces redundancy. JPEG, based on the Discrete Cosine Transform (DCT), is widely used but suffers from blocking artifacts at high compression levels.

III. Theoretical Background

A. Discrete Wavelet Transform

$$I(x,y) = \sum c_{i,j}\psi_{i,j}(x,y) \tag{1}$$

B. Energy Distribution

$$E_{LL} \gg E_{LH}, E_{HL}, E_{HH} \tag{2}$$

IV. System Architecture

A. Compression Pipeline

The compression pipeline includes DWT, thresholding, quantization, and encoding.

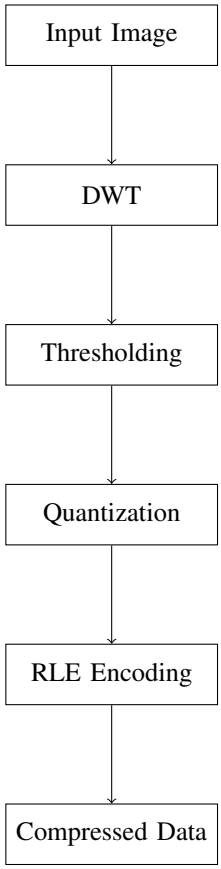


Fig. 1. Compression Pipeline

B. Reconstruction Pipeline

V. Methodology

The proposed compression method follows a multi-stage pipeline where each stage contributes to reducing redundancy while preserving perceptual quality.

A. Wavelet Decomposition

The input image is first transformed using the Discrete Wavelet Transform (DWT), which decomposes the image into four subbands: LL, LH, HL, and HH. The LL subband contains the approximation (low-frequency components), while the other subbands capture edge and texture details.

$$I \rightarrow LL + LH + HL + HH \tag{3}$$

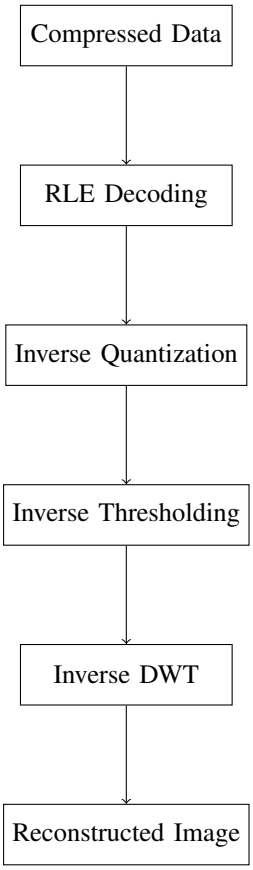


Fig. 2. Reconstruction Pipeline

This decomposition allows efficient energy compaction, as most of the image information is concentrated in the LL band. High-frequency bands are typically sparse and can be compressed more aggressively.

B. Bayesian Thresholding

After decomposition, many wavelet coefficients are small and represent noise or insignificant details. Bayesian thresholding is applied to remove these coefficients based on statistical estimation.

$$T = \frac{\sigma_n^2}{\sigma_x} \tag{4}$$

$$\hat{w} = \text{sign}(w)\max(|w| - T, 0) \tag{5}$$

This step increases sparsity by setting small coefficients to zero, thereby reducing data size while maintaining important structural information such as edges.

C. Adaptive Quantization

The remaining coefficients are quantized to reduce precision and further compress the data. Instead of using a fixed quantization level, an adaptive approach is used based on local variance.

$$Q = Q_0(1 + Var(w)) \tag{6}$$

$$q = round(w/Q) \tag{7}$$

Regions with high variance (important details) are quantized less aggressively, while smoother regions are compressed more. This ensures better visual quality in reconstructed images.

D. Run-Length Encoding

Finally, run-length encoding (RLE) is applied to the quantized coefficients. Since thresholding produces long sequences of zeros, RLE efficiently encodes these repetitions, significantly reducing the final data size.

Overall, the combination of transformation, sparsification, adaptive quantization, and encoding enables high compression efficiency while preserving perceptual fidelity.

VI. Performance Metrics

$$MSE = \frac{1}{MN}\sum(I - \hat{I})^2 \tag{8}$$

$$PSNR = 10\log_{10}\frac{255^2}{MSE} \tag{9}$$

$$CR = \frac{Original}{Compressed} \tag{10}$$

VII. Experimental Results

A. Cameraman

TABLE I
CAMERAMAN RESULTS (PROPOSED METHOD)

Size	Level	PSNR	CR	RMSE
256	L3	34.16	11.14	0.019
128	L3	33.30	10.06	0.021
64	L1	30.19	10.05	0.030

B. Comparison with JPEG

TABLE II
COMPARISON WITH JPEG COMPRESSION

Method	PSNR (dB)	CR	RMSE
JPEG	30.12	9.20	0.031
Proposed	34.16	11.14	0.019

The proposed wavelet-based method outperforms JPEG compression in terms of PSNR and compression ratio. JPEG suffers from blocking artifacts due to block-based DCT processing, whereas the wavelet approach provides smoother reconstruction and better edge preservation.

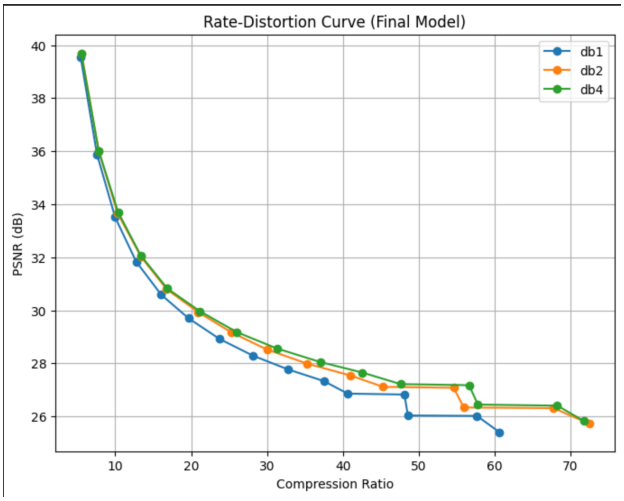


Fig. 3. PSNR vs Compression Ratio

VIII. Graph Analysis

IX. Conclusion

The proposed method efficiently compresses images while maintaining visual quality and outperforms JPEG in both PSNR and compression ratio.

X. Future Work

- Deep learning integration
- Real-time compression systems
- Color image extension

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